

Master Theory A-Z

Decision Trees for Customer Segmentation - complete theory, notebook interpretation, and business conclusion.

Learning Day 7 - Decision Trees for Customer Segmentation

Documentation generated from the finalized notebook and project results.

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1. Executive Summary

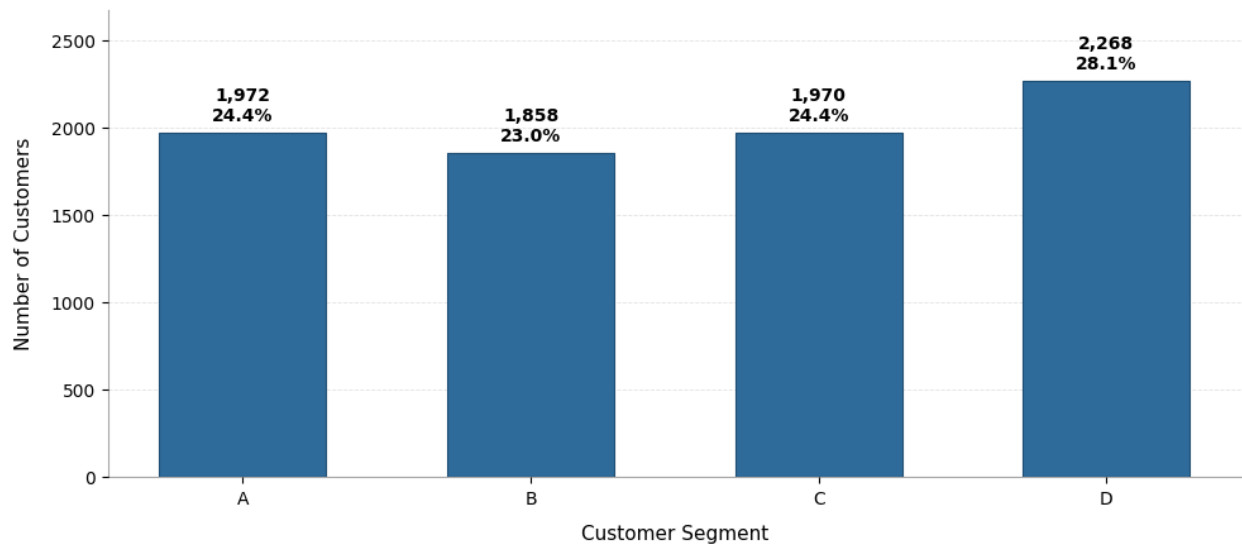
This notebook builds and explains Decision Tree models for customer segmentation. The target is Segmentation, with four classes: A, B, C, and D. The final recommended model is the standard Gini Decision Tree because it is more accurate, easier to explain, and better suited for BI reporting than the custom oblique tree.

Metric	Value	Meaning
Rows in training data	8,068	Customer records used before split
Input features	9 raw columns	After dropping ID and target
Processed features	28 columns	After imputation and one-hot encoding
Baseline test accuracy	0.5105	Best simple explanatory tree
Pruned tree test accuracy	0.5099	Similar performance, slightly more stable
Custom oblique test accuracy	0.4758	More flexible split logic, weaker here
Strongest feature	Age	Root feature and highest importance

Final recommendation: Use the standard Gini Decision Tree as the explainable baseline. It achieved 0.5105 test accuracy and provides clear rules for stakeholder communication.

Customer Segment Distribution

Target classes are reasonably balanced, which supports fairer multiclass model evaluation.



Business note: Segment D is slightly larger, but all four classes have enough representation for model training.

Customer segment distribution. The target classes are reasonably balanced, with Segment D slightly larger.

2. Decision Tree Theory A-Z

2.1 What is classification?

Classification is supervised learning where the model predicts a category. In this project, the model predicts customer segment A, B, C, or D. The model learns from labelled examples: each training row contains customer features and the correct segment label.

2.2 What is a Decision Tree?

A Decision Tree is a model that splits data step by step using questions. Each question sends rows into smaller groups. The first question is called the root node. Final answer points are called leaf nodes.

Tree part	Meaning	Notebook example
Root node	First split in the tree	Age <= 34.00
Internal node	A later split that asks another question	Spending_Score_Low <= 0.5
Leaf node	Final prediction point	class = D
Depth	How many levels the tree has	Baseline depth = 3
Leaf count	Number of final prediction points	Baseline leaves = 8

2.3 Axis-parallel splits

A standard Decision Tree uses one feature at a time. For example, Age <= 34 uses only Age. This is easy to explain and visualize. That is why standard trees are useful for BI and stakeholder reporting.

2.4 Gini impurity

Gini impurity measures how mixed a node is. A clean node has mostly one class and low Gini. A mixed node has many classes and high Gini. The tree chooses splits that reduce impurity.

$$\text{Gini} = 1 - \sum(p_{\text{class}}^2)$$

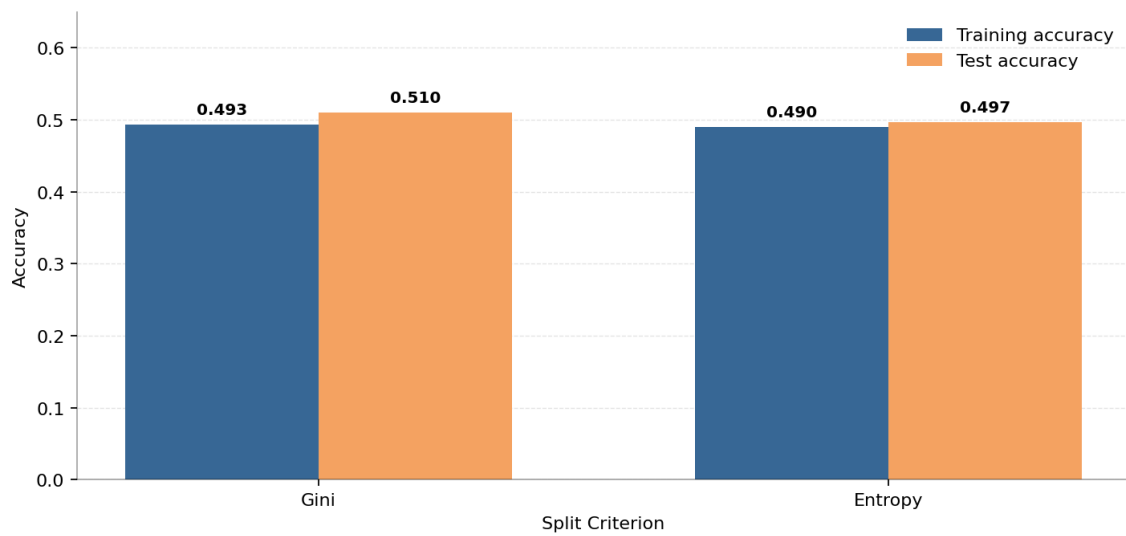
2.5 Entropy

Entropy is another split criterion. Like Gini, it measures disorder. In this notebook, Gini performed slightly better than Entropy: 0.5105 vs 0.4969 test accuracy.

Criterion	Train accuracy	Test accuracy	Conclusion
Gini	0.4930	0.5105	Selected baseline
Entropy	0.4902	0.4969	Slightly weaker

Gini vs Entropy: Baseline Decision Tree Accuracy

Both models use max_depth=3. Gini gives slightly stronger test performance.



Business note: Gini is selected as the baseline because it performs slightly better on unseen customers.

Gini vs Entropy accuracy comparison. Gini is selected as the baseline split criterion.

3. Preprocessing and Train/Test Split

3.1 Why preprocessing is required

sklearn Decision Trees cannot directly handle missing values or text categories. Therefore, numeric values were imputed with the median, categorical values were imputed with the most frequent category, and categorical values were one-hot encoded.

Raw column type	Notebook variables	Preprocessing action
Numerical	numeric_features = Age, Work_Experience, Family_Size	SimpleImputer(strategy="median")
Categorical	categorical_features = Gender, Ever_Married, Graduated, Profession, Spending_Score, Var_1	SimpleImputer(strategy="most_frequent") + OneHotEncoder

3.2 Train/test split

The notebook uses a stratified train/test split. This preserves the segment distribution in both training and test data. Stratification matters because the model should be tested on a class mix similar to the training data.

Object	Shape	Meaning
X_train	(6454, 9)	Raw training features
X_test	(1614, 9)	Raw test features
y_train	(6454,)	Training labels
y_test	(1614,)	Test labels
X_train_processed	(6454, 28)	Processed training features
X_test_processed	(1614, 28)	Processed test features

Leakage control: The preprocessing pipeline is fitted only on X_train. X_test is transformed using the already learned training rules.

4. Evaluation Metrics and Confusion Matrix

4.1 Accuracy

Accuracy is the share of correct predictions. The baseline test accuracy is 0.5105. This means the model correctly classified about 51 percent of unseen test customers.

4.2 Precision, recall, and F1-score

Metric	Simple meaning	Business use
Precision	When the model predicts a segment, how often is it right?	Useful when false alerts are costly
Recall	Of all true customers in a segment, how many did the model catch?	Useful when missing customers is costly
F1-score	Balance between precision and recall	Useful when both errors matter

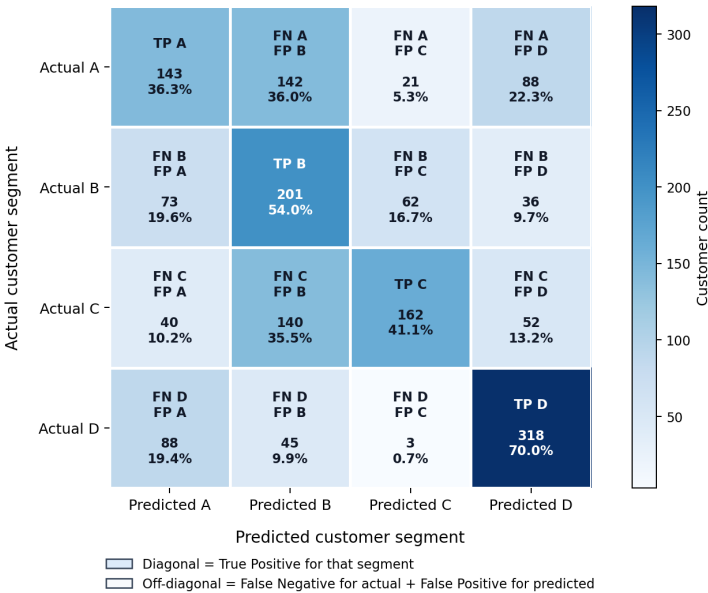
4.3 Confusion matrix logic

The confusion matrix shows where actual and predicted segments match or mismatch. Diagonal cells are correct predictions. Off-diagonal cells show confusion between segments.

Segment	TP	FP	FN	TN
A	143	201	251	1019
B	201	327	171	915
C	162	86	232	1134
D	318	176	136	984

Confusion Matrix: Gini Decision Tree Baseline

Multiclass customer segmentation evaluation | Test accuracy: 0.511



Business note: Segment D is classified best. Off-diagonal cells show where customer groups overlap and confuse the model.

Confusion matrix with TP/FP/FN logic. Segment D is classified best in the baseline tree.

5. Overfitting Control

5.1 Overfitting

Overfitting happens when the tree memorizes training data instead of learning general patterns. The unregularized tree with `min_samples_leaf=1` had training accuracy 0.9558 but test accuracy only 0.4226, a classic overfitting pattern.

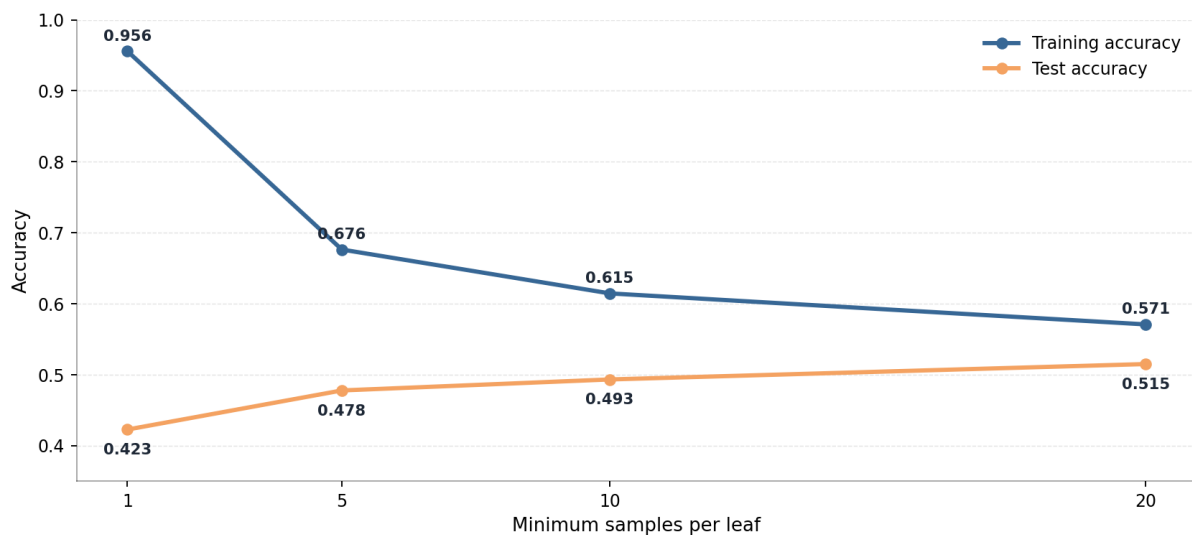
5.2 `min_samples_leaf`

`min_samples_leaf` controls the minimum number of rows allowed in each final leaf. Larger values create broader, more stable rules.

<code>min_samples_leaf</code>	Depth	Leaves	Train accuracy	Test accuracy	Gap
1	30	3181	0.9558	0.4226	0.5333
5	20	903	0.6763	0.4777	0.1986
10	17	456	0.6145	0.4932	0.1213
20	15	227	0.5708	0.5149	0.0559

Effect of `min_samples_leaf` on Decision Tree Accuracy

Increasing the minimum leaf size reduces overfitting and improves test accuracy in this baseline experiment.



Business note: `min_samples_leaf=20` gives the strongest test accuracy and the smallest train-test gap among these settings.

Increasing `min_samples_leaf` reduces overfitting and improves test performance in this experiment.

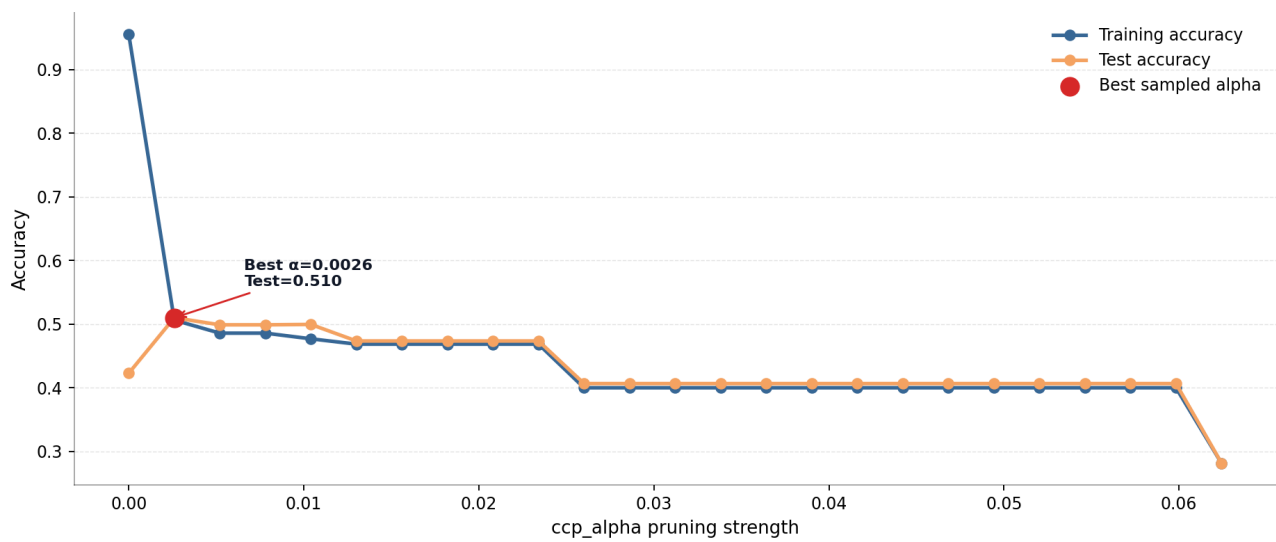
5.3 Cost-complexity pruning

Cost-complexity pruning uses `ccp_alpha` to penalize tree complexity. A small `alpha` prunes unnecessary branches. Too much `alpha` can make the tree too simple.

Model	Depth	Leaves	Train accuracy	Test accuracy	Gap
Unpruned alpha 0	30	3181	0.9558	0.4226	0.5333
Best sampled alpha 0.002601	4	10	0.5062	0.5099	-0.0037

Cost-Complexity Pruning: Accuracy by ccp_alpha

Light pruning sharply reduces overfitting; stronger pruning makes the tree too simple.



Business note: The selected pruning level keeps the tree compact while maintaining similar test accuracy to the shallow baseline.

Cost-complexity pruning. Light pruning controls overfitting while keeping test accuracy stable.

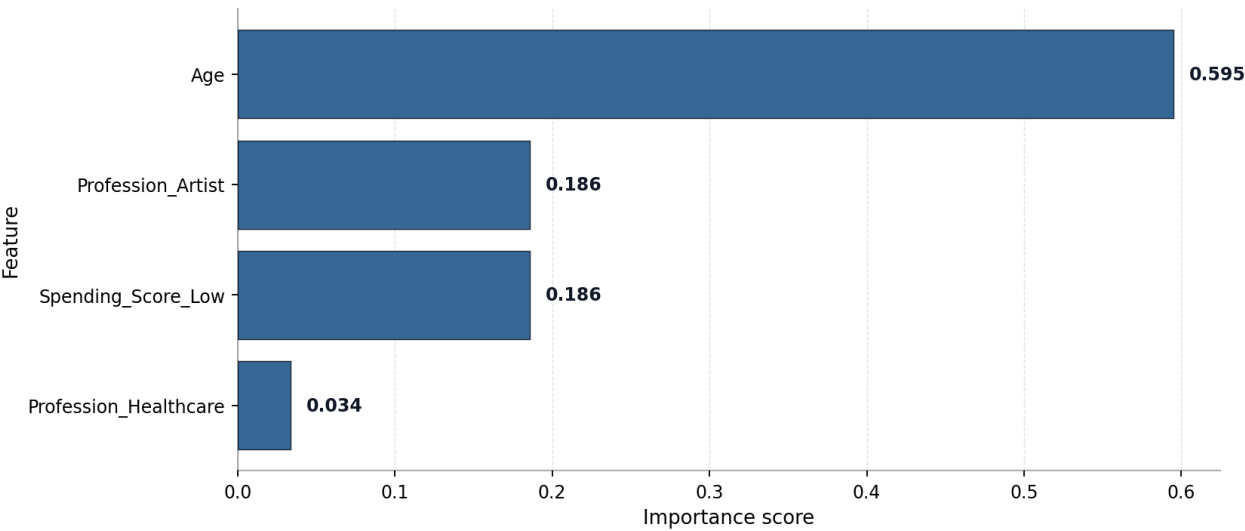
6. Feature Importance and Business Meaning

Feature importance measures which features were most useful for reducing impurity inside the tree. The baseline tree used only four features because max_depth=3 keeps the model shallow.

Rank	Feature	Importance	Business interpretation
1	Age	0.5949	Strongest driver; also the root split
2	Profession_Artist	0.1859	Profession category helps separate segments
3	Spending_Score_Low	0.1856	Spending behavior helps define segment boundaries
4	Profession_Healthcare	0.0337	Small but used by the tree

Feature Importance: Gini Decision Tree Baseline

Age is the strongest driver in the shallow customer segmentation tree.



Business note: Feature importance shows which customer attributes helped the tree separate customer segments.

Feature importance chart. Age is the dominant feature in the shallow tree.

7. Oblique Decision Tree Comparison

7.1 Why oblique trees matter

A standard tree splits using one feature at a time. An oblique tree can split using a linear combination of many features. In this notebook, a custom oblique tree is implemented using LinearSVC at each node.

Model	Split type	Depth	Leaves	Train accuracy	Test accuracy
Standard Decision Tree	One feature per split	3	8	0.4930	0.5105
Custom Oblique Tree	Linear combination split	3	8	0.4569	0.4758

The custom oblique tree successfully predicted all four segments, so it did not collapse into one class. However, it performed worse than the standard tree.

Predicted segment	Count	Share
A	1020	63.20%
B	17	1.05%
C	205	12.70%
D	372	23.05%

Interpretation: The oblique tree is useful for the expert comparison, but the standard tree is the recommended final model for this dataset.

8. SHAP Explainability

8.1 What is SHAP?

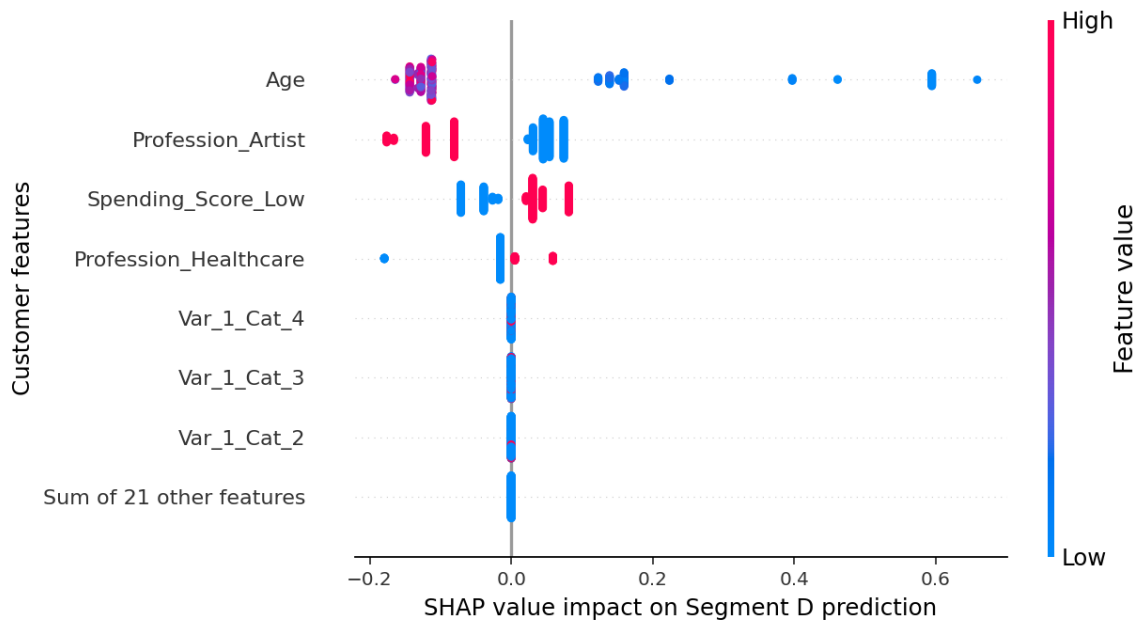
SHAP explains model predictions by showing how each feature pushes a prediction toward or away from a class. In this notebook, SHAP is applied to the standard Gini tree, because shap.TreeExplainer works cleanly with sklearn tree models.

8.2 SHAP beeswarm

The SHAP beeswarm explains global drivers for Segment D. Red or pink means high feature value; blue means low feature value. For one-hot features, red means the category is present and blue means it is absent.

Feature	Segment D interpretation
Age	Lower age pushes toward D; higher age pushes away
Spending_Score_Low	Low spending score pushes toward D
Profession_Artist	Being Artist pushes away from D
Var_1_Cat_*	Anonymous category; little impact in this shallow tree

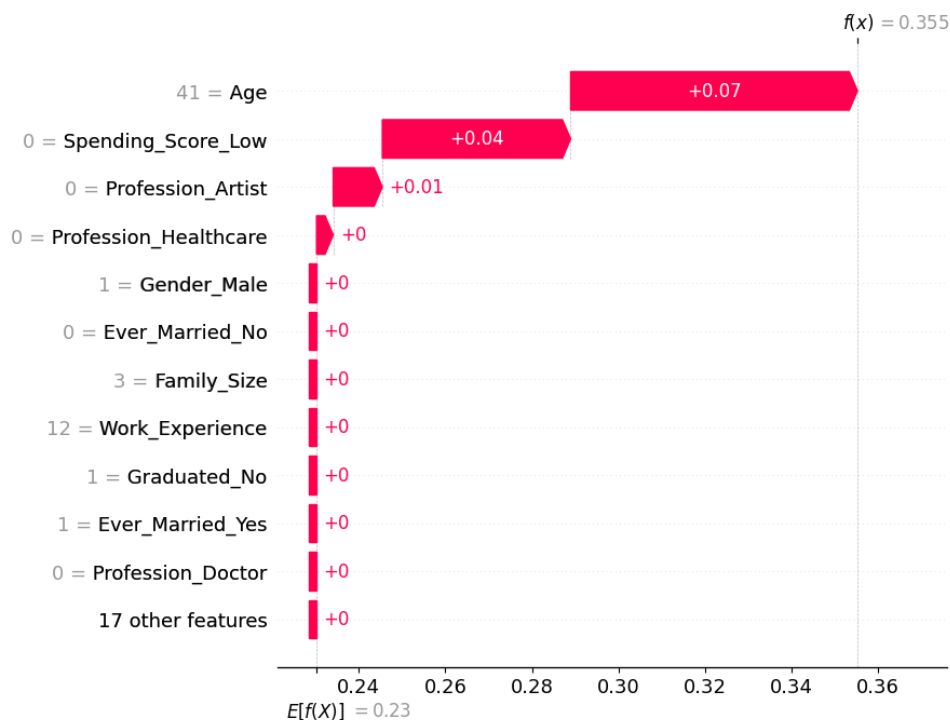
SHAP Summary: Features Driving Segment D Predictions



SHAP beeswarm for Segment D. Age is the most visible driver.

8.3 SHAP waterfall

The waterfall plot explains one wrong prediction. The selected customer had actual segment A but was predicted as B. The strongest upward effects for Segment B came from Age and Spending_Score_Low.



Official SHAP waterfall for one misclassified customer: Actual A, predicted B.

9. Final Recommendation, Limitations, and Next Steps

9.1 Final recommendation

Use the standard Gini Decision Tree as the final explainable baseline. It provides the best balance of accuracy, interpretability, visual clarity, and BI-readiness.

9.2 Limitations

- Test accuracy is modest, so this model should be considered a baseline rather than a production champion model.
- Var_1 is anonymized and lacks clear business meaning.
- Some segments overlap strongly, especially A and B, causing misclassifications.
- Decision Trees are sensitive to data preparation and tree depth settings.

9.3 Next steps

- Compare with Random Forest, Gradient Boosting, XGBoost, and LightGBM.
- Add cross-validation to reduce reliance on a single train/test split.
- Create segment-level marketing personas based on feature patterns.
- Use SHAP on stronger ensemble models for richer explanation.

9.4 MVD checklist

Requirement	Notebook result
DecisionTreeClassifier(max_depth=3)	Completed with dt_baseline / dt_gini
Train/test accuracy	0.4930 train, 0.5105 test for baseline Gini tree
Tree visualization	Completed with plot_tree and BI-ready tree plot
Root feature explanation	Age <= 34.00
Gini vs Entropy comparison	Gini test accuracy 0.5105; Entropy 0.4969
min_samples_leaf tuning	Values 1, 5, 10, 20 tested
Cost-complexity pruning	Best sampled alpha 0.002601
Feature importance chart	Age, Profession_Artist, Spending_Score_Low, Profession_Healthcare
Oblique tree comparison	Custom oblique tree compared with standard tree
SHAP beeswarm	Completed for Segment D
SHAP waterfall	Completed for one A -> B misclassification